TrustRank AI

Intelligent Candidate Discovery & Ranking Engine

Redrob Intelligent Candidate Discovery & Ranking Hackathon

A Multi-Dimensional Scoring Engine and Trap Filter

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Platform: 8-Core CPU (16GB RAM) | Runtime: ~22s for 100K candidates

The Core Recruiting Problem

Limitations of Keyword-Based ATS

- Keyword Stuffing: Candidates list hundreds of AI keywords (LLM, NLP, PyTorch) without real expertise.

- Outsourcing / Consulting Trap: Heavy keyword matching ranks outsourced maintenance roles above product-focused R&D profiles.

- Candidate Availability: Traditional systems ignore critical behavioral signals (activity recency, recruiter response rates).

- Resume Honeypots: Embedded 'trap' profiles with impossible experience data (e.g. 5 yrs skill use in a 3 yr career).

Finding a Founding AI Engineer

- Product over Research: Focus on candidates with practical shipping experience rather than academic publications.

- Fast Trajectory: Spot candidates with rapid growth (promotions, leadership) within small-to-medium product companies.

- Active Engagement: High login activity, high recruiter response rate (>50%), and quick assessment completions.

- Location & Setting: Finding talent aligned with Bangalore/hybrid and ready to work in high-speed settings.

TrustRank AI Architecture

We built a multi-dimensional heuristic scoring pipeline that evaluates candidates across 6 distinct weighted dimensions. Raw JSONL profiles are streamed and scored on-the-fly, generating a composite alignment index in sub-minute speeds.

1. Skill Match (30%)

Evaluates canonical AI skills mapped from 40+ synonyms, weighted by proficiency, duration, endorsements, and tests.

2. Career Trajectory (30%)

Measures title relevance, company pedigree, tenure stability, and applies a 0.4x consulting-firm penalty.

3. Experience Fit (12%)

Uses a Gaussian-like bell curve centered at the ideal 6-8 year sweet-spot, avoiding binary dropoffs.

4. Education (7%)

Scores field relevance (CS/ML/Stats) and matches against Tier-1/Tier-2 engineering institution databases.

5. Behavioral Signals (15%)

Decays score based on login inactivity, recruiter response rate, assessment completion, and notice period.

6. GitHub Activity (6%)

Validates candidate quality using open-source contribution metrics (commits, stars, and PRs).

Dimension Focus: Trust-Weighted Skills

Semantic Synonym Mapping

Skill lists are noisy and unstandardized. We implemented a 40+ synonym map resolving terms to clean taxonomic buckets:

- NLP: sentence-transformers, BERT, SBERT, NLTK, spaCy
- GenAI: LLMs, LangChain, RAG, prompt-engineering
- Deep Learning: PyTorch, TensorFlow, keras, neural-networks
- Core vs Secondary: NLP and Information Retrieval are weighted heavily (1.0). CV and Audio are secondary (0.35).

The Skill-Trust Formula

Instead of counting skills, we compute a Trust Score for each skill entry. Candidates who keyword-stuff with zero duration/endorsements receive negligible scores:

- Proficiency Weight: Beginner = 0.25, Intermediate = 0.5, Advanced = 0.8, Expert = 1.0.

- Endorsement Trust: Logarithmic scaling based on peer endorsements: log(1 + endorsements) / log(50).

- Duration Trust: Linear scaling up to 5 years (60 months) of hands-on experience.

- Assessment Multiplier: Candidates who passed platform assessments receive a 1.25x trust boost.

Dimension Focus: Trajectory & Trap Protection

Career Trajectory & Pedigree

- Title Relevance: Recency-weighted parsing matches roles like 'Senior AI Engineer' (1.0) down to 'HR Manager' (0.0).

- The Consulting Penalty: Entire careers spent at IT services/outsourcing firms (TCS, Infosys, Wipro, Accenture) trigger a severe 0.4x penalty.

- Product Rebound: The consulting penalty is released proportionally if the candidate transitioned into a product company.

- Job-Hopper Check: Average tenure under 12 months triggers a 0.7x penalty on the career score.

Trap & Honeypot Protection

- Honeypot Profiles: The dataset contains ~80 impossible resumes. We inspect dates to find impossible tenure timelines.

- Temporal Validation: If a candidate has worked at a company longer than it has existed, their profile is flagged.

- Skill-Stuffer Validation: If a candidate's skill duration exceeds their total professional experience, they are flagged.

- Hard Disqualification: All flagged profiles receive a hard cap of 0.0 composite score, keeping them out of the top-100 entirely.

Performance & Verification Results

Compute & Efficiency Metrics

- Throughput: Processed 100,000 candidates in 21.7 seconds (4,600+ candidates/second).

- RAM Usage: Under 100MB of RAM, utilizing memory-efficient JSON line streaming.

- Zero GPU / Network: Ranks candidates completely offline without API dependencies or GPU hardware.

- Scalability: Designed to scale to million-row databases with ease on standard CPU instances.

Compliance & shortlisting

- Submission Validity: 100% compliant with the challenge rules (unique ranks 1-100, unique candidate IDs, non-increasing scores).

- Deterministic Tie-Breaking: In case of identical rounded scores, candidate IDs are sorted in ascending order.

- Honeypot Rate: 0.0% honeypots in the top-100 shortlist (disqualification rate is >10%).

- AI Reasoning: Shortlist includes a non-templated, factual, profile-derived rationale for every ranked row.